

HAOCHEN TIAN

Ph.D. Student ◊ NLPR · MAIS · CASIA

Email: tianhaochen2023@ia.ac.cn ◊ GitHub: github.com/hctian713 ◊ [Google Scholar](#)

RESEARCH INTERESTS

Physical AI End-to-End Policy, World Model, Autonomous Driving, Robotics
MultiModal AI Unified Model, Vision-Language Model, Alignment, Evaluation

EDUCATION

Institute of Automation, Chinese Academy of Sciences Sep. 2023 - Present
Ph.D. in Computer Science & Technology, Advisor Prof. [Liang Wang](#), [Tieniu Tan](#)

Wuhan University Sep. 2019 - Jun. 2023
B.S. in Spatial Informatics & Digital Technology

SELECTED PUBLICATIONS

* Equal Contribution † Project Lead ✉ Corresponding Author

Physical AI

SimScale: Learning to Drive via Real-World Simulation at Scale

Haochen Tian, Tianyu Li, Haochen Liu, Jiazhi Yang, Guang Li, Junli Wang, Yinfeng Gao, Zhang Zhang, Liang Wang, Hangjun Ye, Tieniu Tan, Long Chen, Hongyang Li

CVPR 2026 Oral (Top 0.88%, 141/16092). [Paper](#) | [Web](#) | [Code](#) | [Dataset](#) | [Oral Pre](#)

WorldEngine: Towards the Era of Post-Training for Autonomous Driving

OpenDriveLab, Huawei, and Haochen Tian as Contributor

Tech Report 2026. [Paper](#) | [Web](#) | [Code](#)

OneVL: One-Step Latent Reasoning and Planning with Vision-Language Explanation

Xiaomi Embodied Intelligence Team, and Haochen Tian as Contributor

Tech Report 2026. [Paper](#) | [Web](#) | [Code](#)

PlannerRFT: Reinforcing Diffusion Planners through Closed-Loop and Sample-Efficient Fine-Tuning

Hongchen Li, Tianyu Li, Jiazhi Yang, Haochen Tian, Caojun Wang, Lei Shi, Mingyang Shang, Zengrong Lin, Gaoqiang Wu, Zhihui Hao, Xianpeng Lang, Jia Hu, Hongyang Li

CVPR 2026. [Paper](#) | [Web](#)

Reinforced Refinement with Self-Aware Expansion for End-to-End Autonomous Driving

Haochen Liu, Tianyu Li, Haohan Yang, Li Chen, Caojun Wang, Ke Guo, Haochen Tian, Hongchen Li, Hongyang Li, Chen Lv

TPAMI 2026. [Paper](#)

DriveFine: Refining-Augmented Masked Diffusion VLA for Precise and Robust Driving

Chenxu Dang, Sining Ang, Yongkang Li, Haochen Tian, Jie Wang, Guang Li, Hangjun Ye, Jie Ma, Long Chen, Yan Wang

ECCV 2026. [Paper](#) | [Code](#)

TakeVLA: Learning from Mistakes: Post-Training for Driving VLA with Takeover Data

Yinfeng Gao, Deqing Liu, Qichao Zhang, Yupeng Zheng, Haochen Tian, Guang Li, Hangjun Ye, Long Chen, Da-Wei Ding, Dongbin Zhao

Preprint 2026. [Paper](#)

MultiModal AI

A Survey of Unified Multimodal Understanding and Generation: Advances and Challenges

Yan Yang*, Haochen Tian*[†], Yang Shi*, Wulin Xie*, Yi-Fan Zhang[✉], Yuhao Dong, Yibo Hu, Liang Wang, Ran He, Caifeng Shan, Chaoyou Fu[✉], Tieniu Tan

Preprint 2025. [Paper](#) | [Web](#)

RealUnify: Do Unified Models Truly Benefit from Unification?

Haochen Tian as Core Contributor.

[CVPR 2026](#). [Paper](#) | [Code](#) | [Dataset](#)

MME-RealWorld: Could Your Multimodal LLM Challenge High-Resolution Real-World Scenarios that are Difficult for Humans ?

Yi-Fan Zhang, Huanyu Zhang, **Haochen Tian**, Chaoyou Fu, Shuangqing Zhang, Junfei Wu, Feng Li, Kun Wang, Qingsong Wen, Zhang Zhang, Liang Wang, Rong Jin, Tieniu Tan

[ICLR 2025](#). [Paper](#) | [Web](#) | [Code](#) | [Dataset](#)

MM-RLHF: The Next Step Forward in Multimodal LLM Alignment

Yi-Fan Zhang, Tao Yu, **Haochen Tian**, Chaoyou Fu, Peiyan Li, Jianshu Zeng, Wulin Xie, Yang Shi, Huanyu Zhang, Junkang Wu, Xue Wang, Yibo Hu, Bin Wen, Fan Yang, Zhang Zhang, Tingting Gao, Di Zhang, Liang Wang, Rong Jin, Tieniu Tan

[ICML 2025](#). [Paper](#) | [Web](#) | [Code](#) | [Dataset](#)

BaseReward: A Strong Baseline for Multimodal Reward Model

Yi-Fan Zhang, Haihua Yang, Huanyu Zhang, Yang Shi, Zezhou Chen, **Haochen Tian**, Chaoyou Fu, Haotian Wang, Kai Wu, Bo Cui, Xu Wang, Jianfei Pan, Haotian Wang, Zhang Zhang, Liang Wang

[ICLR 2026](#). [Paper](#)

Scaling and Beyond: Advancing Spatial Reasoning in MLLMs Requires New Recipes

Huanyu Zhang, Chengzu Li, Wenshan Wu, Shaoguang Mao, Yifan Zhang, **Haochen Tian**, Ivan Vulić, Zhang Zhang, Liang Wang, Tieniu Tan, Furu Wei

Preprint 2025. [Paper](#)

How Well Do Models Follow Visual Instructions? VIBE: A Systematic Benchmark for Visual Instruction-Driven Image Editing

Huanyu Zhang, Xuehai Bai, Chengzu Li, Chen Liang, **Haochen Tian**, Haodong Li, Ruichuan An, Yifan Zhang, Anna Korhonen, Zhang Zhang, Liang Wang, Tieniu Tan

Preprint 2026. [Paper](#) | [Web](#) | [Code](#)

EXPERIENCE

Xiaomi EV, Embodied Intelligence Team

*Research Intern, **Top Talent Program***

Sep. 2025 - Present

Beijing, China

- Worked with Prof. [Long Chen](#) and [Naiyan Wang](#)
- Worked on scalable closed-loop training for end-to-end autonomous driving.

Shanghai AI Lab → The University of Hong Kong, OpenDriveLab

Research Intern

Jun. 2024 - Present

Hong Kong SAR, China

- Worked with Prof. [Hongyang Li](#)
- Worked on human feedback reward for end-to-end autonomous driving.

Megvii Research, Foundation Model Group

Research Intern

Feb. 2024 - Mar. 2024

Beijing, China

- Worked on 3D-Aware Vision-Language-Action model for end-to-end autonomous driving.

HONORS / AWARDS

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| • Merit Student of University of Chinese Academy of Sciences | 2024 |
| • Outstanding Graduate of Wuhan University | 2023 |
| • Wangzhizhuo Innovation Scholarship (Top 1%) | 2023 |
| • Gold Award of “Internet+” Innovation Competition | 2022 |
| • Merit Student of Wuhan University | 2021/2022 |

ACADEMIC SERVICE

Competition Organizer

AlpaSim End-to-End Closed-Loop Challenge 2026

Workshop Organizer

CVPR 2026 Workshop on [From Labs to Life: Embodied Intelligence in the Wild](#)

Standards Working Group Member

IEEE Standard P3474: [Human Intentions and AI Alignment in Autonomous Driving Agents](#)

Conference/Journal Reviewer

CVPR, ECCV, NeurIPS, ICRA, RSS, WACV, BMCV, RA-L, T-PAMI, etc.

Invited Talk

Scalable Real-World Simulation Reshapes End-to-End Policy Learning THU | AD Heart | CV3D etc, Dec. 2025

SKILLS

Language

Chinese (Native) | English (Proficient CET6 500+, CET4 600+)

Programming

Python, C++, Matlab, Java, JavaScript, SQL